Predicting the Next Category and Region of a Visit

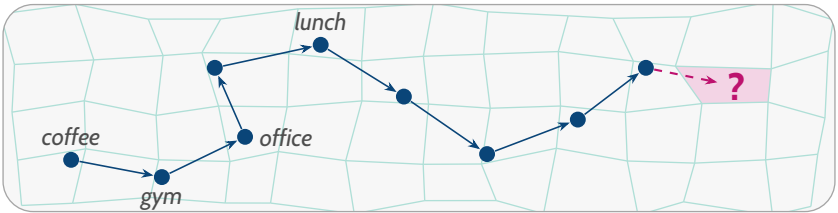
A Check-in-Level Multi-Task Study on Mobility Data

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One person's day, and two questions



What kind of place?

Next category: one of 7 types, such as food, shopping, nightlife. → *what the user will want*

Which neighborhood?

Next region: its census tract (Istanbul: *mahalle*), 520 to 8,501. → *where to prepare*

Check-in: a visit shared on a location-based social network such as Gowalla: a place (a *point of interest*, POI) and a time (Cho 2011). Uses: caching, capacity, demand anticipation (Bastug 2014; Moura 2025). A tract is not a radio cell: we claim nothing about cells or handover.

Two models, or one?

- **Two dedicated models**: one per question; two things to train, version and serve.
- **One joint model**: one artifact to deploy, and the tasks might help each other. But shared parameters can settle on a compromise good for neither task (Caruana 1997).

Multi-task models in this field use category and region as auxiliary signals for next-place prediction (MCMG, Sun 2024; HMT-GRN, Lim 2022). To our knowledge, fine-grained region as an end target of equal standing is underexplored.

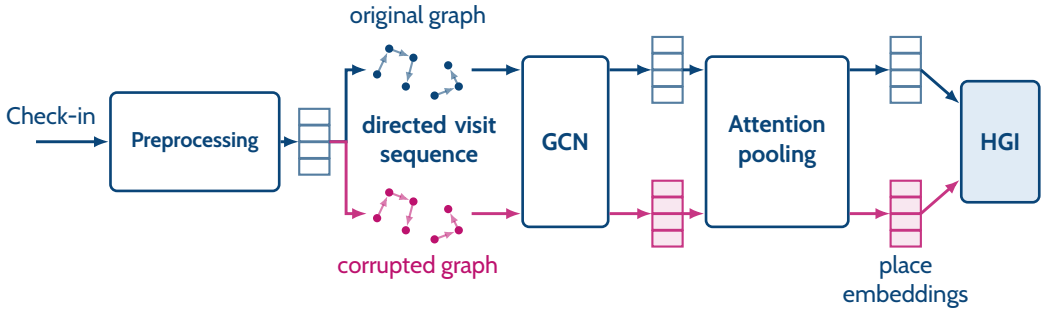
The question

Can one model predict both the category and the region of a user's next visit, and what does it cost?

Short answer: yes, at a measured price.

Check2HGI: a fourth level below the place

Place embeddings: one vector per place, DGI (Veličković 2019), HGI (Huang 2023). Two visits to one café look identical.



- **Each visit carries** its place's category, the hour and day of week, and four elapsed-time values → one **64-d** vector per visit, **the vector our model reads**; pooled per place, it feeds the levels above.
- Trained **without the task labels**, and **built from the past alone**: edges between consecutive visits run forward in time only.

How it was measured

Regions	census tracts (Istanbul: *mahalle*), **520** to **8,501**
Split	by user: the prediction models never train on a test user's visits
Scored	category: **macro-F1** (always-majority: 5.7 to 7.3)
	region: **Acc@10**, is the true region in the top ten? (random: at most about 2%)
Repeated	**20 trained models** per configuration (4 seeds $\times$ 5 folds)
Decided	written down before any result was seen:
	category: is the joint model **better**? (superiority)
	region: is it **not worse** by more than **2 Acc@10 points**? (non-inferiority, TOST) (Lakens 2017)

Result 1: the representation, at every dataset

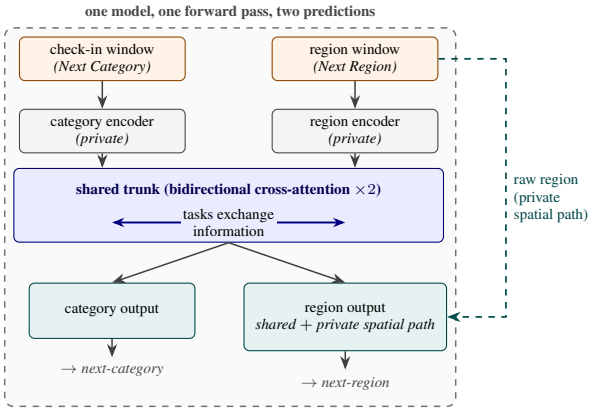
Next category, single-task model, seed 0 — only the input changes

Dataset	Check-in level	Place level (HGI)	Δ
Alabama	**30.77** $\pm$1.16	29.15 $\pm$0.84	$+1.62$
Arizona	**34.51** $\pm$1.15	31.93 $\pm$1.08	$+2.58$
Istanbul	**35.35** $\pm$0.89	32.54 $\pm$1.03	$+2.81$
Florida	**37.36** $\pm$0.42	37.13 $\pm$0.41	$+0.23$
California	**35.62** $\pm$0.47	34.74 $\pm$0.41	$+0.88$
Texas	**36.32** $\pm$0.51	35.33 $\pm$0.48	$+0.99$

macro-F1; five matched folds, $\pm$ is the fold sd.

Larger at the three smaller datasets ($+1.6$ to $+2.8$), under one point at the three largest. A paired test separates the two at five of six; not at Florida.

One model for two tasks



Multi-task learning: one model does several tasks at once by sharing most of its parts.

The risk: what the tasks share can settle on a compromise worse for each.

Here: each task has its own input and encoder. In the shared trunk, attention lets each task read the other's features while keeping its own weights: **they share by exchanging information**.

A private spatial path keeps what is purely spatial out of the exchange.

One model, two answers: the price

	Next region (Acc@10)			**Next category** (macro-F1)		
	Ded.	Joint	Δ	Ded.	Joint	Δ
Alabama	**70.12**	69.24	−0.87	**30.77**	30.59	−0.19
Arizona	**59.48**	59.04	−0.44	**34.57**	**34.57**	−0.00
Istanbul	**75.16**	75.08	−0.08	35.34	**35.42**	+0.08
Florida	**76.69**	76.54	−0.16	37.35	**37.55**	+0.19
California	63.48	**64.54**	+1.06	**35.63**	**35.63**	−0.00
Texas	64.94	**66.15**	+1.21	**36.33**	36.19	−0.13

Ded.: two dedicated models; Joint: one model. **Green**: outperforms (Holm-corrected). Bold: the larger. **Region**: the other four are deficits, not ties, inside the 2-point margin. **Category**: every interval within half a point; only Florida survives correction. Δ: paired mean over four seeds (can differ from the rounded columns).

The trade is a measured one and not a free substitution.

And against published systems

Next category (macro-F1)

	POI-RGNN	Ded.	Joint
AL	23.80	**30.77**	30.59
AZ	27.64	**34.57**	**34.57**
IST	30.12	35.34	**35.42**
FL	33.35	37.35	**37.55**
CA	30.71	**35.63**	**35.63**
TX	32.08	**36.33**	36.19

Next region (Acc@10)

	HMT-GRN	ReHDM	STAN	Ded.	Joint
AL	63.19	65.38	60.72	**70.12**	69.24
AZ	52.11	53.00	49.86	**59.48**	59.04
IST	68.06	69.33	61.86	**75.16**	75.08
FL	72.76	64.49	72.99	**76.69**	76.54
CA	59.59	50.26	58.61	63.48	**64.54**
TX	61.39	48.81	61.75	64.94	**66.15**

Both our models are above every system we compared against; the joint model by at least **4.11** (category) and **3.55** (region). That margin includes the representation's advantage, so it is not evidence for sharing. Bold: best in row; underline: second.

POI-RGNN (Capanema 2023): next category, re-implemented. **HMT-GRN** (Lim 2022): multi-task next place + regions, next-place parts removed; our windows and folds, best epoch; not the full published system. **STAN** (Luo 2021): re-implemented, its own representations and splits, ranks regions. **ReHDM** (Li 2025): its own split.

What a mobility-aware service would receive

For example, a service that caches content where users are heading, or provisions capacity before demand arrives (Bastug 2014), at the level of a neighborhood.

- **Time to act** — half of the next visits come within **0.4 h** (Florida) to **5.5 h** (Istanbul) of the previous one; 5 to 27% come more than 3 days later.
- **A shortlist** — ask the model for its ten most likely regions. In California (8,501) the true one is among them **64.54%** of the time; in Texas (6,553), **66.15%**. This is the Acc@10 of slide 7, read as a hit rate.
- **For scale** — a transition table with no learning reaches 51 to 72 Acc@10; the joint model is 4 to 10 points above it everywhere.

We built and evaluated no such service: this is motivation, not a measured service result.

What it costs, and three limits

THE COST

Parameters, joint vs **the two dedicated models combined**:

AL 4.2 M vs 1.9 M
CA 5.2 M vs 2.8 M

One pass costs **more** compute than the two small models together. The gain is operational: one artifact to train, version and deploy; two answers.

THREE LIMITS

1. Representation trained **once over all users**; a per-fold rebuild moved the dedicated models by at most 0.17 macro-F1 and 0.13 Acc@10 (one seed, the datasets checked).
2. Epoch selection reads the scored fold: **absolute scores are optimistic**; joint vs dedicated is affected far less.
3. The joint model is **larger**, so its region gains at Texas and California may come from size, not sharing. The one-model result does not depend on which.

Conclusion

- **Built**: a check-in-level representation that extends hierarchical graph infomax from the place to the individual visit, a fixed per-visit vector that other models can reuse; and one model that answers both questions.
- **Measured**: on region, the joint model outperforms the dedicated model at Texas and California and stays within the 2-point margin elsewhere (deficits, not ties); on category, within half a point. **Above every published system we compared against.** A measured trade, not a free substitution.
- **Next**: extend the joint model to the **exact next place**, as a third target on the same representation; an inductive variant for unseen places and users; newer, denser data and more cities; how far a miss lands, inside a service.

What a user will do next, and where, from one model instead of two.

Thank you

Questions?

Code (model, representation, baselines, statistical tests)
`github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac`

Data (public) · category-annotated Gowalla · Massive-STEPS (Istanbul)

Supported by **FAPEMIG**, **CNPq** and **CAPES**.

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B1 · Why census tracts

- Predicting over a partition of the map is standard, usually with a **grid cell**; we use **official neighborhood-scale units**;
- Next region is classification over all the dataset's regions:

	Istanbul	AL	AZ	FL	TX	CA
Regions	520	1,109	1,547	4,703	6,553	8,501

B2 · The margin, and the tests

- **Registered before any result was read**: superiority for category, non-inferiority for region with a margin of **2 Acc@10 points**, paired Wilcoxon;
- **Why 2**: a service that acts on which neighborhood will be busy would not behave differently below it; the seed-to-seed sd of the paired difference is 0.02 to 0.16;
- **Reported**: paired t on the four per-seed means, 90% interval, Holm across the six datasets per task; at four seeds the exact Wilcoxon cannot go below 0.0625. Both tests are in the code release;
- Category has **no registered equivalence margin**: a difference that fails superiority is unresolved, never a match;
- The two region gains are **secondary results**, outside the registered plan.

B3 · The Markov floor

- A first-order Markov chain over region transitions, no learning: **51.23 to 72.47** Acc@10;
- The joint model exceeds it by $+4.1$ **to** $+10.0$ at every dataset; the narrowest is **Florida,** $+4.07$.

B4 · Can a visit vector see its target?

- Each visit node draws on the **visits that precede it**: the vector encodes the visit with its own history;
- The graph does **not pass information from a later visit back to an earlier one**, in training or at readout;
- That is what keeps a node from carrying a feature of the target it is used to predict;
- The graph is trained without the task labels: it never sees the next category or the next region.

B5 · One model instead of two

- **4.2 M vs 1.9 M** parameters at Alabama, **5.2 M vs 2.8 M** at California, against **the two dedicated models combined**;
- One pass costs more compute than running the two small models;
- The gain is operational: one artifact to train, version and deploy; one pass returns both answers.

B6 · Then what is multi-task learning for?

- On category, the representation moves the score **more than the choice between one model and two**;
- Whether the exchange between the tasks adds anything beyond the representation is not separated by the evidence here;
- What one model gives for certain is operational: one artifact, two answers.

B7 · The external systems

- **HMT-GRN** (Lim 2022) — primary region comparison: our windows and folds, one seed, best validation epoch; its multi-task structure kept, the graph components and hierarchical beam search removed, a per-fold region-transition prior added (our own reported models do not use it); not the complete published system;
- **STAN** (Luo 2021) — re-implemented with its own representations and splits from the raw check-ins, output adapted to rank regions, seed 0;
- **ReHDM** (Li 2025) — its own published chronological split; single seed at TX and CA;
- **POI-RGNN** (Capanema 2023) — next category, re-implemented, non-overlapping windows.

B8 · More regions, more gain?

	Regions	Region Δ (Acc@10)
Texas	6,553	$+1.21$
California	8,501	$+1.06$

California has more regions than Texas and a **smaller** gain.

An observation, not a law.

B9 · Is it just the raw features?

Control: the place vector + the same raw per-visit features (category one-hot, hour, day of week), same single-task model. Run at three datasets.

	Control gain	Check-in-level gain
Alabama	$+1.73$	$+1.62$
Arizona	$+1.70$	$+2.58$
Florida	$+1.02$	$+0.23$

macro-F1 over the place-level representation, category axis.

"Most of the category difference is therefore already available in the raw per-visit features, and this control does not separate what the check-in-level representation adds beyond them on this axis."

B10 · The intervals

Joint minus dedicated, 90% interval of the paired difference

	Next region (Acc@10)		**Next category** (macro-F1)	
	Δ	90% CI	Δ	90% CI
Alabama	-0.87	$[-1.00, -0.75]$	-0.19	$[-0.33, -0.04]$
Arizona	-0.44	$[-0.62, -0.25]$	-0.00	$[-0.04, +0.03]$
Istanbul	-0.08	$[-0.16, -0.002]$	$+0.08$	$[+0.01, +0.15]$
Florida	-0.16	$[-0.19, -0.13]$	$+0.19$	$[+0.14, +0.25]$
California	$+1.06$	$[+1.03, +1.08]$	-0.00	$[-0.03, +0.02]$
Texas	$+1.21$	$[+1.13, +1.29]$	-0.13	$[-0.19, -0.08]$

Green: outperforms, Holm-corrected — region at Texas ($p = 0.00013$) and California ($p < 10^{-4}$), 20 of 20 folds; category at Florida ($p = 0.011$), 19 of 20.

Every region interval of the other four lies **entirely below zero**: deficits, not ties (Istanbul only just).

B11 · References

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B12 · The datasets

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Dens. (%)	Maj. (%)
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

Lengths: check-ins per user. Density: $100\times$ check-ins / (users $\times$ POIs). Majority: share of next-visit labels in the most common category. Check-ins, users and POIs count the full corpus; windows are counted after the ten-visit user filter.